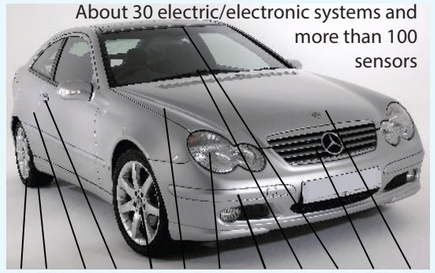
Technology Brief 7: Resistive Sensors

An ***electrical sensor*** is a device capable of responding to an applied ***stimulus*** by generating an electrical signal whose voltage, current, or some other attribute is related to the intensity of the stimulus.

► The family of possible stimuli encompasses a wide array of physical, chemical, and biological quantities, including temperature, pressure, position, distance, motion, velocity, acceleration, concentration (of a gas or liquid), blood flow, etc. ◄

The sensing process relies on measuring resistance, capacitance, inductance, induced electromotive force (emf), oscillation frequency or time delay, among others. Sensors are integral to the operation of just about every instrument that uses electronic systems, from automobiles and airplanes to computers and cell phones (**Fig. TF7-1**). This Technology Brief covers resistive sensors. ***Capacitive***, ***inductive***, and ***emf sensors*** are covered separately (here and in later chapters).



System	Abbrev.	Sensors
Distronic	DTR	3
Electronic controlled transmission	ECT	9
Roof control unit	RCU	7
Antilock braking system	ABS	4
Central locking system	ZV	3
Dyn. beam levelling	LWR	6
Common-rail diesel injection	CDI	11
Automatic air condition	AAC	13
Active body control	ABC	12
Tire pressure monitoring	TPM	11
Electron. stability program	ESP	14
Parktronic system	PTS	12

Figure TF7-1 Most cars use on the order of 100 sensors.

Piezoresistivity

According to Eq. (4.70), the resistance of a cylindrical resistor or wire conductor is given by $R = l/\sigma A$, where l is the cylinder's length, A is its cross-sectional area, and σ is the conductivity of its material. Stretching the wire by an applied external force causes l to increase and A to decrease. Consequently, R increases (**Fig. TF7-2**). Conversely, compressing the wire causes R to decrease. The Greek word ***piezein*** means to press, from which the term piezoresistivity is derived. *This should not be confused with piezoelectricity, which is an emf effect. (See EMF Sensors in Technology Brief 12.)*

The relationship between the resistance R of a piezoresistor and the ***applied force*** F can be modeled by the approximate linear equation

$$R = R_0\left(1 + \frac{\alpha F}{A_0}\right),$$

where R_0 is the unstressed resistance (@ $F = 0$), A_0 is the unstressed cross-sectional area of the resistor, and α is the ***piezoresistive coefficient*** of the resistor material. The force F is positive if it is causing the resistor to stretch and negative if it is compressing it.

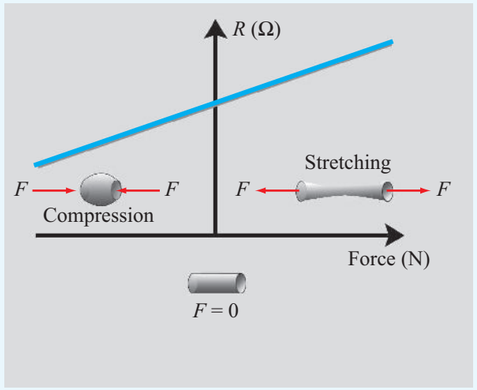


Figure TF7-2 Piezoresistance varies with applied force.

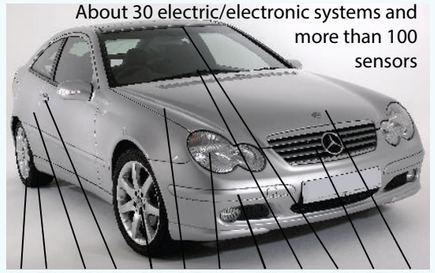
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技术简介 7：电阻传感器

电传感器是一种能够通过生成电信号来响应施加的刺激的设备，该电信号的电压、电流或一些其他属性与刺激的强度相关。

► 可能的刺激家族涵盖了广泛的物理、化学和生物量，包括温度、压力、位置、距离、运动、速度、加速度、（气体或液体）浓度、血流等。◄

传感过程依赖于测量电阻、电容、电感、感应电动势 (emf)、振荡频率或时间延迟等。从汽车和飞机到计算机和手机，几乎所有使用电子系统的仪器的操作都离不开传感器（图 TF7-1）。本技术简介涵盖电阻传感器。电容式、电感式和电动势传感器分别介绍（此处和后面的章节）。



System	Abbrev.	Sensors
Distronic	DTR	3
Electronic controlled transmission	ECT	9
Roof control unit	RCU	7
Antilock braking system	ABS	4
Central locking system	ZV	3
Dyn. beam levelling	LWR	6
Common-rail diesel injection	CDI	11
Automatic air condition	AAC	13
Active body control	ABC	12
Tire pressure monitoring	TPM	11
Electron. stability program	ESP	14
Parktronic system	PTS	12

图 TF7-1 大多数汽车使用大约 100 个传感器。

压阻率

根据方程。根据方程（4.70），圆柱形电阻器或导线的电阻由 R = l/σ A 给出，其中 l 是圆柱体的长度，A 是其横截面积，σ 是其材料的电导率。通过施加外力拉伸导线会导致 l 增加而 A 减少。因此，R 增加（图 TF7-2）。相反，压缩导线会导致 R 减小。希腊语“piezein”的意思是“按压”，“压阻率”一词就是由此而来。这不应与压电相混淆，压电是一种电动势效应。（请参阅技术简介 12 中的 EMF 传感器。）

压敏电阻的电阻 R 与施加的力 F 之间的关系可以通过近似线性方程来建模

$$R = R_0\left(1 + \frac{\alpha F}{A_0}\right),$$

其中 R_0 是无应力电阻 (@ F = 0)，A_0 是电阻器的无应力横截面积，α 是电阻器材料的压阻系数。如果力 F 导致电阻器拉伸，则力 F 为正；如果力 F 压缩电阻器，则力 F 为负。

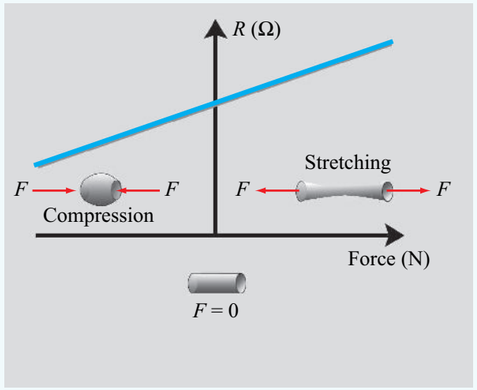


图 TF7-2 压阻随施加的力而变化。

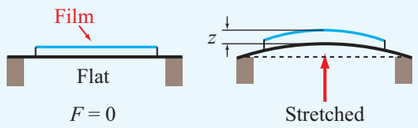
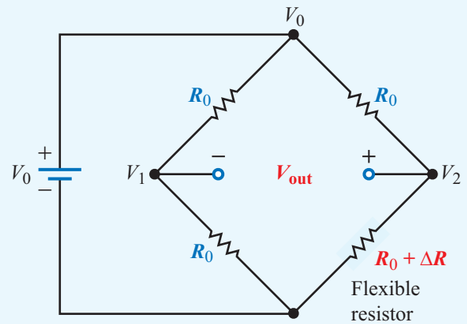


Figure TF7-3 Piezoresistor films.

► By connecting the piezoresistor to a Wheatstone bridge circuit (**Fig. TF7-5**) where the other three resistors are all identical in value and equal to R_0 (the resistance of the piezoresistor when no external force is present), the voltage output becomes directly proportional to the normalized resistance change: $\Delta R/R_0$. ◄

An elastic resistive sensor is well suited for measuring the deformation z of a surface (**Fig. TF7-3**), which can be related to the pressure applied to the surface; and if z is recorded as a function of time, it is possible to derive the velocity and acceleration of the surface's motion. To realize high longitudinal piezoresistive sensitivity (the ratio of the normalized change in resistance, $\Delta R/R_0$, to the corresponding change in length, $\Delta l/l_0$, caused by the applied force), the piezoresistor is often designed as a serpentine-shaped wire (**Fig. TF7-4(a)**) bonded on a flexible plastic substrate and glued onto the surface whose deformation is to be monitored. Copper and nickel alloys are commonly used for making the sensor wires, although in some applications silicon is used instead (**Fig. TF7-4(b)**) because it has a very high piezoresistive sensitivity.



$$V_{\text{out}} = \frac{V_0}{4}\left(\frac{\Delta R}{R_0}\right)$$

Figure TF7-5 Wheatstone bridge circuit with piezoresistor.

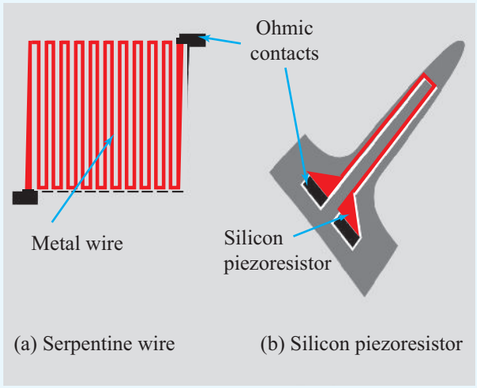


Figure TF7-4 Metal and silicon piezoresistors.

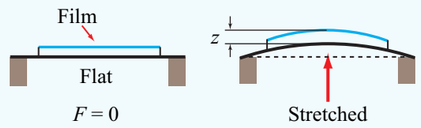
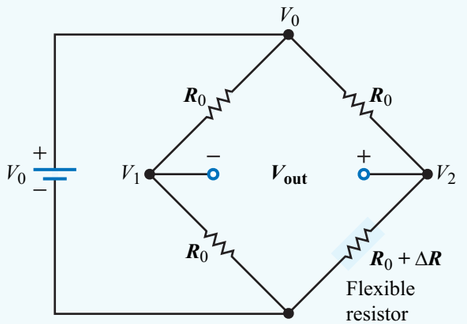


图 TF7-3 压敏电阻薄膜。

► 通过将压敏电阻连接到惠斯通电桥电路（图 TF7-5），其中其他三个电阻的阻值都相同，并且等于 R_0（无外力时压敏电阻的电阻），电压输出与归一化电阻变化成正比：$\Delta R/R_0$。◄

弹性电阻传感器非常适合测量表面的变形 z（图 TF7-3），这可能与施加到表面的压力有关；如果将 z 记录为时间的函数，则可以推导出表面运动的速度和加速度。为了实现高纵向压阻灵敏度（归一化电阻变化 $\Delta R/R_0$ 与由施加的力引起的相应长度变化 $\Delta l/l_0$ 的比率），压敏电阻通常设计为蛇形线（图 TF7-4(a)），粘合在柔性塑料基板上，并粘合到要监测变形的表面上。铜和镍合金通常用于制造传感器线，尽管在某些应用中使用硅代替（图TF7-4（b）），因为它具有非常高的压阻灵敏度。



$$V_{\text{out}} = \frac{V_0}{4}\left(\frac{\Delta R}{R_0}\right)$$

图 TF7-5 带压敏电阻的惠斯登电桥电路。

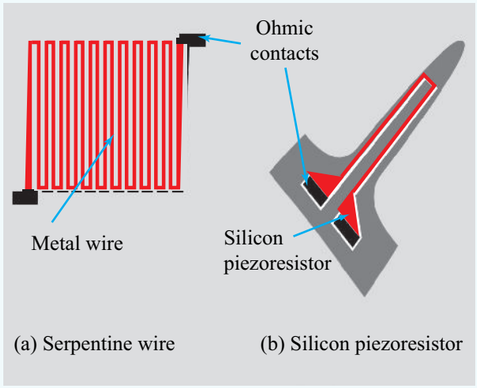


图 TF7-4 金属和硅压敏电阻。

4-5.2 Electric Potential Due to Point Charges

The electric field due to a point charge q located at the origin is given by Eq. (4.32) as

$$\mathbf{E} = \hat{\mathbf{R}}\,\frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}. \tag{4.54}$$

The field is radially directed and decays quadratically with the distance R from the observer to the charge.

As was stated earlier, the choice of integration path between the end points in Eq. (4.43) is arbitrary. Hence, we can conveniently choose the path to be along the radial direction $\hat{\mathbf{R}}$, in which case $d\mathbf{l} = \hat{\mathbf{R}}\,dR$ and

$$V = -\int_{\infty}^{R}\left(\hat{\mathbf{R}}\,\frac{q}{4\pi\varepsilon R^2}\right)\cdot\hat{\mathbf{R}}\,dR = \frac{q}{4\pi\varepsilon R} \quad \text{(V)}. \tag{4.55}$$

If the charge q is at a location other than the origin, say at position vector $\mathbf{R}_1$, then V at observation position vector $\mathbf{R}$ becomes

$$V = \frac{q}{4\pi\varepsilon|\mathbf{R}-\mathbf{R}_1|} \quad \text{(V)}, \tag{4.56}$$

where $|\mathbf{R}-\mathbf{R}_1|$ is the distance between the observation point and the location of the charge q. The principle of superposition applied previously to the electric field $\mathbf{E}$ also applies to the electric potential V. Hence, for N discrete point charges $q_1, q_2, \ldots, q_N$ residing at position vectors $\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_N$, the electric potential is

$$V = \frac{1}{4\pi\varepsilon}\sum_{i=1}^{N}\frac{q_i}{|\mathbf{R}-\mathbf{R}_i|} \quad \text{(V)}. \tag{4.57}$$

4-5.3 Electric Potential Due to Continuous Distributions

To obtain expressions for the electric potential V due to continuous charge distributions over a volume υ', across a surface S', or along a line l', we (1) replace q_i in Eq. (4.57) with $\rho_v\,d\upsilon'$, $\rho_s\,ds'$, and $\rho_\ell\,dl'$, respectively; (2) convert the summation into an integration; and (3) define $R' = |\mathbf{R}-\mathbf{R}_i|$ as the distance between the integration point and the observation point. These steps lead to the following expressions:

$$V = \frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\rho_v}{R'}\,d\upsilon' \quad \textbf{(volume distribution)}, \tag{4.58a}$$

$$V = \frac{1}{4\pi\varepsilon}\int_{S'}\frac{\rho_s}{R'}\,ds' \quad \textbf{(surface distribution)}, \tag{4.58b}$$

$$V = \frac{1}{4\pi\varepsilon}\int_{l'}\frac{\rho_\ell}{R'}\,dl' \quad \textbf{(line distribution)}. \tag{4.58c}$$

4-5.4 Electric Field as a Function of Electric Potential

In Section 4-5.1, we expressed V in terms of a line integral over $\mathbf{E}$. Now we explore the inverse relationship by reexamining Eq. (4.37):

$$dV = -\mathbf{E}\cdot d\mathbf{l}. \tag{4.59}$$

For a scalar function V, Eq. (3.73) gives

$$dV = \nabla V\cdot d\mathbf{l}, \tag{4.60}$$

where ∇V is the gradient of V. Comparison of Eq. (4.59) with Eq. (4.60) leads to

$$\mathbf{E} = -\nabla V. \tag{4.61}$$

▶ This differential relationship between V and $\mathbf{E}$ allows us to determine $\mathbf{E}$ for any charge distribution by first calculating V and then taking the negative gradient of V to find $\mathbf{E}$. ◀

The expressions for V, given by Eqs. (4.57) to (4.58c), involve scalar sums and scalar integrals, and as such are usually much easier to evaluate than the vector sums and integrals in the expressions for $\mathbf{E}$ derived in Section 4-3 on the basis of Coulomb's law. Thus, even though the electric potential approach for finding $\mathbf{E}$ is a two-step process, it is conceptually and computationally simpler to apply than the direct method based on Coulomb's law.

Example 4-9: Electric Field of an Electric Dipole

An ***electric dipole*** consists of two point charges of equal magnitude but opposite polarity separated by a distance d (**Fig. 4-15(a)**). Determine V and $\mathbf{E}$ at any point P given that P is at a distance $R \gg d$ from the dipole center and the dipole resides in free space.

Solution: To simplify the derivation, we align the dipole along the z axis and center it at the origin (**Fig. 4-15(a)**). For the two charges shown in **Fig. 4-15(a)**, application of Eq. (4.57) gives

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{R_1}+\frac{-q}{R_2}\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{R_2-R_1}{R_1R_2}\right).$$

Since $d \ll R$, the lines labeled R_1 and R_2 in **Fig. 4-15(a)** are approximately parallel to each other, in which case the following approximations apply:

$$R_2 - R_1 \approx d\cos\theta, \qquad R_1R_2 \approx R^2.$$

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4-5.2 点电荷产生的电势

由位于原点的点电荷 q 产生的电场由方程 1 给出。 (4.32) 作为

$$\mathbf{E} = \hat{\mathbf{R}}\,\frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}. \tag{4.54}$$

该场呈放射状，并随观察者到电荷的距离 R 呈二次方衰减。

如前所述，方程中端点之间积分路径的选择。 (4.43) 是任意的。因此，我们可以方便地选择沿径向方向 R̂ 的路径，在这种情况下 dl = R̂ dR 和

$$V = -\int_{\infty}^{R}\left(\hat{\mathbf{R}}\,\frac{q}{4\pi\varepsilon R^2}\right)\cdot\hat{\mathbf{R}}\,dR = \frac{q}{4\pi\varepsilon R} \quad \text{(V)}. \tag{4.55}$$

如果电荷 q 位于原点以外的位置，例如位置矢量 R_1，则观测位置矢量 R 处的 V 变为

$$V = \frac{q}{4\pi\varepsilon|\mathbf{R}-\mathbf{R}_1|} \quad \text{(V)}, \tag{4.56}$$

其中 $|R - R_1|$ 是观察点与电荷 q 位置之间的距离。先前应用于电场 E 的叠加原理也适用于电势 V 。因此，对于位于位置向量 R_1、R_2、…、R_N 处的 N 个离散点电荷 q_1、q_2、…、q_N，电势为

$$V = \frac{1}{4\pi\varepsilon}\sum_{i=1}^{N}\frac{q_i}{|\mathbf{R}-\mathbf{R}_i|} \quad \text{(V)}. \tag{4.57}$$

4-5.3 连续分布产生的电势

为了获得由于体积 υ′、表面 S′或沿线 l′的连续电荷分布而产生的电势 V 的表达式，我们 (1) 替换方程 1 中的 q_i。 (4.57) 分别为 $\rho_v\,d\upsilon'$、$\rho_s\,ds'$ 和 $\rho_\ell\,dl'$； (2) 将求和转化为积分； (3)定义$R' = |R-R_i|$为积分点与观测点之间的距离。这些步骤导致以下表达式：

$$V = \frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\rho_v}{R'}\,d\upsilon' \quad \textbf{(volume distribution)}, \tag{4.58a}$$

$$V = \frac{1}{4\pi\varepsilon}\int_{S'}\frac{\rho_s}{R'}\,ds' \quad \textbf{(surface distribution)}, \tag{4.58b}$$

$$V = \frac{1}{4\pi\varepsilon}\int_{l'}\frac{\rho_\ell}{R'}\,dl' \quad \textbf{(line distribution)}. \tag{4.58c}$$

4-5.4 电场作为电势的函数

在第 4-5.1 节中，我们用 E 上的线积分来表示 V。现在我们通过重新检查方程来探索逆关系。 (4.37) ：

$$dV = -\mathbf{E}\cdot d\mathbf{l}. \tag{4.59}$$

对于标量函数 V ， 方程： (3.73) 给出

$$dV = \nabla V\cdot d\mathbf{l}, \tag{4.60}$$

其中 ∇V 是 V 的梯度。等式的比较(4.59) 与等式。 (4.60) 导致

$$\mathbf{E} = -\nabla V. \tag{4.61}$$

► V 和 E 之间的这种微分关系使我们能够通过首先计算 V，然后采用 V 的负梯度来找到 E，从而确定任何电荷分布的 E。◄

V 的表达式由等式给出。 (4.57) 到 (4.58c) 涉及标量和和标量积分，因此通常比第 4-3 节中根据库仑定律导出的 E 表达式中的向量和和积分更容易计算。因此，尽管寻找 E 的电势方法是一个两步过程，但它在概念上和计算上都比基于库仑定律的直接方法更容易应用。

例 4-9：电偶极子的电场

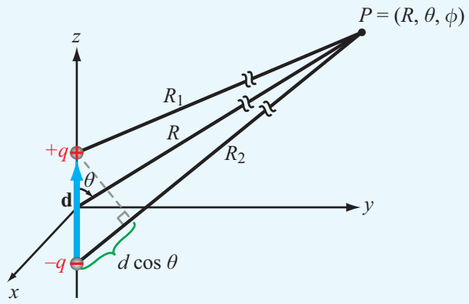
电偶极子由两个大小相等但极性相反、间隔距离 d 的点电荷组成（图 4-15(a)）。假设 P 距偶极子中心的距离为 R ≫ d 并且偶极子位于自由空间中，请确定任意点 P 处的 V 和 E。

解：为了简化推导，我们将偶极子沿z轴对齐，并以原点为中心（图4-15（a））。对于图 4-15(a) 所示的两种电荷，应用等式： (4.57) 给出

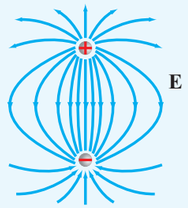
$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{R_1}+\frac{-q}{R_2}\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{R_2-R_1}{R_1R_2}\right).$$

由于 d ≪ R，图 4-15(a) 中标记为 R_1 和 R_2 的线近似彼此平行，在这种情况下，适用以下近似：

$$R_2 - R_1 \approx d\cos\theta, \qquad R_1R_2 \approx R^2.$$



(a) Electric dipole



(b) Electric-field pattern

Figure 4-15 Electric dipole with dipole moment $\mathbf{p} = q\mathbf{d}$ (Example 4-9).

Hence,

$$V = \frac{qd\cos\theta}{4\pi\varepsilon_0 R^2}. \tag{4.62}$$

To generalize this result to an arbitrarily oriented dipole, note that the numerator of Eq. (4.62) can be expressed as the dot product of $q\mathbf{d}$ (where $\mathbf{d}$ is the distance vector from $-q$ to $+q$) and the unit vector $\hat{\mathbf{R}}$ pointing from the center of the dipole toward the observation point P. That is,

$$qd\cos\theta = q\mathbf{d}\cdot\hat{\mathbf{R}} = \mathbf{p}\cdot\hat{\mathbf{R}}, \tag{4.63}$$

where $\mathbf{p} = q\mathbf{d}$ is called the ***dipole moment***. Using Eq. (4.63) in Eq. (4.62) then gives

$$V = \frac{\mathbf{p}\cdot\hat{\mathbf{R}}}{4\pi\varepsilon_0 R^2} \quad \textbf{(electric dipole)}. \tag{4.64}$$

In spherical coordinates, Eq. (4.61) is given by

$$\mathbf{E} = -\nabla V = -\left(\hat{\mathbf{R}}\frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial V}{\partial\theta} + \hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial V}{\partial\phi}\right), \tag{4.65}$$

where we have used the expression for ∇V in spherical coordinates given in Appendix C. Upon taking the derivatives of the expression for V given by Eq. (4.62) with respect to R and θ and then substituting the results in Eq. (4.65), we obtain

$$\mathbf{E} = \frac{qd}{4\pi\varepsilon_0 R^3}\left(\hat{\mathbf{R}}2\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta\right) \quad \text{(V/m)}. \tag{4.66}$$

We stress that the expressions for V and $\mathbf{E}$ given by Eqs. (4.64) and (4.66) apply only when $R \gg d$. To compute V and $\mathbf{E}$ at points in the vicinity of the two dipole charges, it is necessary to perform all calculations without resorting to the far-distance approximations that led to Eq. (4.62). Such an exact calculation for $\mathbf{E}$ leads to the field pattern shown in **Fig. 4-15(b)**.

4-5.5 Poisson's Equation

With $\mathbf{D} = \varepsilon\mathbf{E}$, the differential form of Gauss's law given by Eq. (4.26) may be cast as

$$\nabla\cdot\mathbf{E} = \frac{\rho_v}{\varepsilon}. \tag{4.67}$$

Inserting Eq. (4.61) in Eq. (4.67) gives

$$\nabla\cdot(\nabla V) = -\frac{\rho_v}{\varepsilon}. \tag{4.68}$$

Given Eq. (3.110) for the Laplacian of a scalar function V,

$$\nabla^2 V = \nabla\cdot(\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}, \tag{4.69}$$

Eq. (4.68) can be cast in the abbreviated form

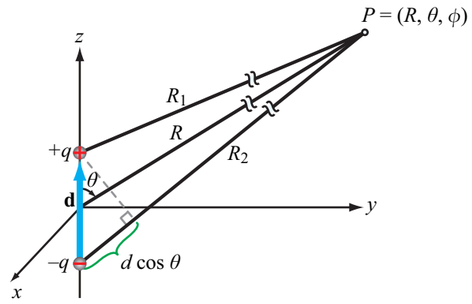
$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad \textbf{(Poisson's equation)}. \tag{4.70}$$

This is known as ***Poisson's equation***. For a volume υ' containing a volume charge density distribution ρ_v, the solution for V derived previously and expressed by Eq. (4.58a) as

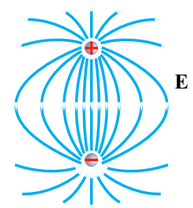
$$V = \frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\rho_v}{R'}\,d\upsilon' \tag{4.71}$$

satisfies Eq. (4.70). If the medium under consideration contains no charges, Eq. (4.70) reduces to

$$\nabla^2 V = 0 \quad \textbf{(Laplace's equation)}, \tag{4.72}$$



(a) Electric dipole



(b) Electric-field pattern

图 4-15 偶极矩为 p = qd 的电偶极子（示例 4-9）。

因此，

$$V = \frac{qd\cos\theta}{4\pi\varepsilon_0 R^2}. \tag{4.62}$$

为了将此结果推广到任意方向的偶极子，请注意方程的分子。 (4.62) 可以表示为 qd（其中 d 是从 −q 到 +q 的距离向量）与从偶极子中心指向观测点 P 的单位向量 R̂ 的点积。即

$$qd\cos\theta = q\mathbf{d}\cdot\hat{\mathbf{R}} = \mathbf{p}\cdot\hat{\mathbf{R}}, \tag{4.63}$$

其中 p = qd 称为偶极矩。使用方程式方程（4.63） (4.62) 然后给出

$$V = \frac{\mathbf{p}\cdot\hat{\mathbf{R}}}{4\pi\varepsilon_0 R^2} \quad \textbf{(electric dipole)}. \tag{4.64}$$

在球坐标系中，方程(4.61) 由下式给出

$$\mathbf{E} = -\nabla V = -\left(\hat{\mathbf{R}}\frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial V}{\partial\theta} + \hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial V}{\partial\phi}\right), \tag{4.65}$$

其中我们使用了附录 C 中给出的球坐标中的 ∇V 表达式 。 （4.62）关于R和θ，然后将结果代入式（4.62）。 （4 .65），我们得到

$$\mathbf{E} = \frac{qd}{4\pi\varepsilon_0 R^3}\left(\hat{\mathbf{R}}2\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta\right) \quad \text{(V/m)}. \tag{4.66}$$

我们强调 V 和 E 的表达式由等式给出。 (4.64) 和 (4.66) 仅当 R ≫ d 时适用。为了计算两个偶极子电荷附近点处的 V 和 E，有必要执行所有计算，而无需诉诸导致方程 1 的远距离近似。 （4.62）。对 E 的这种精确计算得出如图 4-15(b) 所示的场模式。

4-5.5 泊松方程

对于 D = ε E，高斯定律的微分形式由方程给出。 (4.26) 可以表示为

$$\nabla\cdot\mathbf{E} = \frac{\rho_v}{\varepsilon}. \tag{4.67}$$

插入方程式方程（4.61） (4.67) 给出

$$\nabla\cdot(\nabla V) = -\frac{\rho_v}{\varepsilon}. \tag{4.68}$$

给定方程。 (3.110) 对于标量函数 V 的拉普拉斯算子，

$$\nabla^2 V = \nabla\cdot(\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}, \tag{4.69}$$

等式。 (4.68) 可以缩写为形式

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad \textbf{(Poisson's equation)}. \tag{4.70}$$

这就是所谓的泊松方程。对于包含体积电荷密度分布 ρ_v 的体积 υ′，之前导出的 V 的解由方程 1 表示。 （4.58a ）为

$$V = \frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\rho_v}{R'}\,d\upsilon' \tag{4.71}$$

满足方程。 （4.70）。如果所考虑的介质不包含电荷，则方程： (4.70) 减少为

$$\nabla^2 V = 0 \quad \textbf{(Laplace's equation)}, \tag{4.72}$$

and it is then referred to as ***Laplace's equation***. Poisson's and Laplace's equations are useful for determining the electrostatic potential V in regions with boundaries on which V is known, such as the region between the plates of a capacitor with a specified voltage difference across it.

Concept Question 4-9: What is a conservative field?

Concept Question 4-10: Why is the electric potential at a point in space always defined relative to the potential at some reference point?

Concept Question 4-11: Explain why Eq. (4.40) is a mathematical statement of Kirchhoff's voltage law.

Concept Question 4-12: Why is it usually easier to compute V for a given charge distribution and then find $\mathbf{E}$ using $\mathbf{E} = -\nabla V$ than to compute $\mathbf{E}$ directly by applying Coulomb's law?

Concept Question 4-13: What is an electric dipole?

Exercise 4-10: Determine the electric potential at the origin due to four 20 μC charges residing in free space at the corners of a 2 m $\times$ 2 m square centered about the origin in the x–y plane.

Answer: $V = \sqrt{2} \times 10^{-5}/(\pi\varepsilon_0)$ (V). (See EM.)

Exercise 4-11: A spherical shell of radius a has a uniform surface charge density ρ_s. Determine (a) the electric potential and (b) the electric field with both at the center of the shell.

Answer: (a) $V = \rho_s a/\varepsilon$ (V), (b) $\mathbf{E} = 0$. (See EM.)

4-6 Conductors

The electromagnetic ***constitutive parameters*** of a material medium are its electrical permittivity ε, magnetic permeability μ, and conductivity σ. A material is said to be ***homogeneous*** if its constitutive parameters do not vary from point to point, and ***isotropic*** if they are independent of direction. Most materials are isotropic, but some crystals are not. Throughout this book, all materials are assumed to be homogeneous and isotropic. This section is concerned with σ, Section 4-7 examines ε, and discussion of μ is deferred to Chapter 5.

► The ***conductivity*** of a material is a measure of how easily electrons can travel through the material under the influence of an externally applied electric field. ◄

Materials are classified as ***conductors*** (metals) or ***dielectrics*** (insulators) according to the magnitudes of their conductivities. A conductor has a large number of loosely attached electrons in the outermost shells of its atoms. In the absence of an external electric field, these free electrons move in random directions and with varying speeds. Their random motion produces zero average current through the conductor. Upon applying an external electric field, however, the electrons migrate from one atom to the next in the direction opposite that of the external field. Their movement gives rise to a ***conduction current density***

$$\mathbf{J} = \sigma\mathbf{E} \quad (\text{A/m}^2) \quad \textbf{(Ohm's law)}, \tag{4.73}$$

where σ is the material's conductivity with units of siemen per meter (S/m).

In yet other materials, called dielectrics, the electrons are tightly bound to the atoms—so much so that it is very difficult to detach them under the influence of an electric field. Consequently, no significant conduction current can flow through them.

► A ***perfect dielectric*** is a material with $\sigma = 0$. In contrast, a ***perfect conductor*** is a material with $\sigma = \infty$. Some materials, called superconductors, exhibit such a behavior. ◄

The conductivity σ of most metals is in the range from 10^6 to 10^7 S/m when compared with 10^{-10} to 10^{-17} S/m for good insulators (**Table 4-1** on p. 194). A class of materials called ***semiconductors*** allow for conduction currents even though their conductivities are much smaller than those of metals. The conductivity of pure germanium, for example, is 2.2 S/m. Tabulated values of σ at room temperature (20 °C) are given in Appendix B for some common materials, and a subset is reproduced in **Table 4-1**.

► The conductivity of a material depends on several factors, including temperature and the presence of impurities. In general, σ of metals increases with decreasing temperature. Most superconductors operate in the neighborhood of absolute zero. ◄

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然后它被称为拉普拉斯方程。泊松方程和拉普拉斯方程可用于确定具有 V 已知边界的区域中的静电电势 V，例如具有指定电压差的电容器极板之间的区域。

概念问题 4-9：什么是保守域？

概念问题 4-10：为什么空间中某一点的电势总是相对于某个参考点的电势来定义？

概念问题 4-11：解释为什么(4.40) 是基尔霍夫电压定律的数学表述。

概念问题 4-12：为什么计算给定电荷分布的 V，然后使用 E = −∇V 求 E，通常比应用库仑定律直接计算 E 更容易？

概念问题 4-13：什么是电偶极子？

练习 4-10：确定由于位于 x-y 平面中以原点为中心的 2 m × 2 m 正方形的角上的自由空间中的四个 20 μ C 电荷而在原点处的电势。

Answer: $V = \sqrt{2} \times 10^{-5}/(\pi\varepsilon_0)$ (V). (See EM.)

练习4-11：半径为a的球壳具有均匀的表面电荷密度ρ_s。确定 (a) 电势和 (b) 电场（两者均位于壳的中心）。

答案：(a) V = ρ_sa/ε (V)，(b) E = 0。（参见EM。）

4-6 导体

材料介质的电磁本构参数是其介电常数ε、磁导率μ和电导率σ。如果一种材料的本构参数在点与点之间不发生变化，则该材料被称为均质材料；如果材料的本构参数与方向无关，则该材料被称为各向同性材料。大多数材料是各向同性的，但有些晶体不是。在本书中，所有材料都被假设为均匀且各向同性的。本节关注 σ，第 4-7 节研究 ε，对 μ 的讨论推迟到第 5 章。

► 材料的电导率是电子在外部施加的电场的影响下穿过材料的容易程度的量度。 ◄

根据其电导率的大小，材料被分为导体（金属）或电介质（绝缘体）。导体的原子最外层有大量松散连接的电子。在没有外部电场的情况下，这些自由电子以随机方向和不同速度移动。它们的随机运动产生通过导体的平均电流为零。然而，一旦施加外部电场，电子就会以与外部电场相反的方向从一个原子迁移到下一个原子。它们的运动产生传导电流密度

$$\mathbf{J} = \sigma\mathbf{E} \quad (\text{A/m}^2) \quad \textbf{(Ohm's law)}, \tag{4.73}$$

其中 σ 是材料的电导率，单位为西门子每米 (S/m)。

在另一些称为电介质的材料中，电子与原子紧密结合，以至于在电场的影响下很难将它们分离。因此，没有明显的传导电流可以流过它们。

► 理想的电介质是 σ = 0 的材料。相反，理想的导体是 σ = ∞ 的材料。一些称为超导体的材料表现出这种行为。 ◄

大多数金属的电导率 σ 范围为 10^6 至 10^7 S/m，而良好绝缘体的电导率为 10^{-10} 至 10^{-17} S/m（第 194 页的表 4-1）。一类称为半导体的材料允许传导电流，尽管它们的电导率比金属小得多。例如，纯锗的电导率为 2.2 S/m。附录 B 中给出了一些常见材料在室温 (20 °C) 下的 σ 表格值，表 4-1 中再现了一个子集。

► 材料的电导率取决于多种因素，包括温度和杂质的存在。一般来说，金属的σ随着温度的降低而增加。大多数超导体在绝对零附近运行。 ◄